As we mentioned in the opening paragraphs of this chapter, Elasticsearch provides APIs to index both single and multiple documents into Elasticsearch. We go over these document APIs in detail in the next section.

5.1.1 Document APIs

We communicate with Elasticsearch using RESTful APIs over HTTP. We can perform basic CRUD (create, read, update, and delete) operations using single-document APIs. There are also APIs designed to work with multiple documents instead of targeting a single document. We discuss both types in this chapter, but for now, let's focus on how to index documents using single-document APIs. To begin, there's an important concept to understand: document identifiers.

Document identifiers

Each document we index can have an identifier (ID), usually specified by the user. For example, the document for the movie *The Godfather* can be given an ID (say, `id = 1`), and the movie *The Shawshank Redemption* can be given another ID (`id = 2`). Similar to a primary key in a relational database, the ID is associated with that document for its lifetime (unless it is deliberately changed).

Sometimes the client (user) doesn't need to give a document an ID. Imagine an automatic car sending thousands of alerts and heartbeats to a server. Not every message needs a sequence of IDs; it can have a random ID, albeit a unique one. In this case, the system generates a random universally unique identifier (UUID) for the document that's being indexed.

Document APIs allow us to index documents with or without an ID. But there is a subtle difference when using HTTP verbs such as `POST` and `PUT`:

- If a document has an ID provided by the client, we use the HTTP `PUT` method to invoke a document API for indexing the document.
- If the document does not have an ID provided by the client, we use the HTTP `POST` method when indexing. In this case, once the document is indexed, it inherits the system-generated ID.

Let's check out the process of indexing documents using both methods.

Indexing a document with an identifier (PUT)

When a document has an ID, we can use the single-document index (`_doc`) API with HTTP's `PUT` action to index the document. The syntax for this method is as follows:

```
PUT <index_name>/_doc/<identifier>
```

Here, `<index_name>` is the name of the index where the document will be housed, and `_doc` is the endpoint that must be present when indexing a document. The `<identifier>` is the document's identity (like a primary key in a database), which is a mandatory path parameter when using the HTTP `PUT` method.

Let's index a `movie` document using the API. Head to Kibana, and write and execute the following query.